

SPINE SPECIFICATION

Admissibility-Controlled Epistemic Governance for Machine Reasoning

1. Field of the Invention

The present invention relates to artificial intelligence and machine reasoning systems. More specifically, it relates to architectures that govern whether a system is permitted to produce a determination, commitment, or resolution, particularly in the presence of ambiguity, underspecified context, or competing interpretations.

The invention is concerned with **epistemic governance**, not language generation, belief modeling, or probabilistic prediction.

2. Background

2.1 The Technical Problem

Contemporary artificial reasoning systems exhibit a structural failure mode in which outputs are generated despite insufficient information to justify resolution. This failure manifests as:

- premature resolution of ambiguous inputs,
- forced selection among multiple valid interpretations,
- post-hoc rationalization of unsupported conclusions, and
- inability to terminate processing with a correct non-answer.

These behaviors arise not from incorrect training data, but from architectural constraints that require systems to always produce an answer and that optimize for plausibility, coherence, or confidence rather than admissibility.

Such systems lack mechanisms to determine when a decision itself is **not permitted**.

2.2 Limitations of Existing Approaches

Existing approaches to epistemic reasoning typically operate by forming, revising, or ranking beliefs, propositions, or truth-valued assertions. In such systems:

- ambiguity is treated as uncertainty or insufficient evidence,
- suspension of belief is temporary and oriented toward eventual resolution, and
- non-commitment is represented as absence rather than as a governed outcome.

These approaches govern *what may be believed*, but do not govern *whether a determination may be made at all*. As a result, they do not prevent premature epistemic collapse.

3. Summary of the Invention (The Spine)

The invention provides a **machine reasoning system that enforces epistemic governance by controlling admissibility of commitment**.

The system:

- explicitly represents ambiguity as a persistent epistemic state,
- maintains multiple valid interpretations concurrently,
- prohibits collapse or commitment unless structural constraints force resolution, and
- produces explicit refusal or clarification requests as correct terminal outputs when admissibility conditions are not met.

Commitment is not a belief update and does not represent acceptance of a proposition. Commitment is a **constrained state transition** permitted only when admissibility conditions are satisfied.

Where such conditions are not satisfied, the system deliberately **refuses to decide**.

This governance is architectural, deterministic, and auditable.

As a technical effect, the invention deterministically prevents hallucinated or unjustified commitments by prohibiting resolution in the absence of admissibility-satisfying structural constraints.

4. Non-Belief-Based Epistemic Governance (Core Constraint)

In one embodiment, the invention explicitly does **not** operate by forming, revising, ranking, or maintaining beliefs, belief bases, propositional commitments, truth-values, or belief acceptance thresholds.

The system does not perform belief revision, belief suspension, truth-maintenance, probabilistic justification, or normative belief evaluation.

Instead, the system governs **epistemic admissibility** as a control property of the reasoning process itself.

Interpretations, hypotheses, or candidate outcomes are represented as **epistemic states**, not beliefs.

Commitment is permitted only as a structurally forced transition between epistemic states.

Refusal constitutes correct system behavior and does not represent uncertainty, lack of confidence, deferred belief formation, or probabilistic indecision.

5. System Architecture (Functional Overview)

5.1 Epistemic State Representation

The system maintains an explicit representation of epistemic states, including:

- unresolved ambiguity states,
- resolved states, and
- inadmissible states.

Multiple epistemic states may coexist concurrently without ranking or preference.

5.2 Admissibility Gate

An admissibility gate evaluates whether a transition from an unresolved state to a committed state is permitted.

The gate enforces structural constraints derived from context, bindings, and system rules.

No commitment may occur without passing this gate.

In some embodiments, the admissibility gate produces an explicit admissibility state indicator, which may be exposed to an external interface or downstream system. Such exposure does not constitute commitment or resolution, but represents a governed epistemic state. In other embodiments, the same admissibility state is enforced silently without external signalling. Both forms are within the scope of the invention.

5.3 Collapse Controller

A collapse controller performs commitment only when admissibility conditions are satisfied.

Collapse is forbidden in the absence of forcing constraints.

Heuristic resolution, probabilistic weighting, or narrative completion are explicitly prohibited.

5.4 Refusal Output Module

Where admissibility conditions are not met, the system terminates processing with an explicit refusal or clarification request.

Refusal is treated as a valid terminal outcome.

Upon refusal, the system terminates the reasoning process for the queried determination without producing an alternative resolution.

Where clarification is produced, the clarification is constrained to admissible bindings or determinations derivable from existing epistemic states. The system does not solicit unrestricted or speculative input, and does not expand scope beyond the original determination.

6. Epistemic State Dynamics

6.1 Ambiguity Persistence

Ambiguity is represented as a stable epistemic state.

There is no requirement that ambiguity resolve over time.

The system is permitted to remain indefinitely in an unresolved state.

6.2 Constraint-Forced Collapse

Collapse is permitted only when:

- competing interpretations are eliminated by structural constraints, or
- binding information forces uniqueness.

Absent such constraints, collapse is forbidden.

6.3 Refusal Semantics

Refusal prevents hallucinated commitment and preserves system integrity.

Refusal is not a failure state and does not require follow-up inference.

7. Embodiments (Non-Limiting)

7.1 Compositional Primitive Embodiment

In one embodiment, epistemic governance is implemented via explicit compositional primitives that define:

- activation preconditions,
- state updates, and
- collapse conditions.

7.2 Structured State Embodiment

In one embodiment, epistemic states are structured using roles, domains, and spans.

7.3 Persistent Existence Frame Embodiment

In one embodiment, epistemic states operate within a non-temporal present-state frame in which persistence does not depend on timeline indexing.

This embodiment is optional and non-limiting.

8. Governed Comparison and Evaluation Protocol (Optional Embodiment)

In one embodiment, the system evaluates external references, claims, or assertions using a governed claim-by-claim comparison protocol that:

- forbids narrative inference,
- requires structural element matching, and
- permits refusal as a terminal outcome where determination is underdetermined.

9. Advantages

The invention:

- prevents premature epistemic collapse by construction,
- enables correct non-answers,
- ensures determinism and auditability, and
- may be integrated with external language generation systems as non-authoritative expression layers without delegating epistemic authority to them.

10. Priority and Incorporation

This specification may claim priority from earlier disclosures relating to:

- persistent epistemic state substrates,
- compositional reasoning primitives, and
- structured conceptual architectures.

Such disclosures are incorporated as embodiments and do not limit the scope of the invention.

Appendix A — Mapping of Prior Provisional Disclosures to the Present Specification

Mapping of Existing Provisionals to Spine Specification

Spine Reference Anchor

Spine Document:

Admissibility-Controlled Epistemic Governance for Machine Reasoning

Core Invariant of the Spine:

Epistemic admissibility governs whether a system is permitted to decide; ambiguity is persistent unless structurally forced to collapse; refusal is a correct terminal outcome.

Everything maps *into* that invariant.

1. Persistent Existence Frame (PEF) Provisional

(Temporal / substrate invention)

What PEF Claims (at its core)

- Cognition occurs in a continuous present state
- Persistence does not require timeline indexing
- Past and future are reconstructed/projected, not lived
- State continuity exists without belief accumulation

Where PEF Maps into the Spine

Primary Spine Sections:

- §6.1 Ambiguity Persistence
- §7.3 Persistent Existence Frame Embodiment
- §4 Non-Belief-Based Epistemic Governance (supporting logic)

How the Mapping Works (explicitly)

Spine Requirement	PEF Contribution
Ambiguity may persist indefinitely	PEF removes temporal pressure toward resolution
No obligation to “eventually decide”	Present-state cognition has no future-forcing

Spine Requirement

Refusal can be terminal

Parallel epistemic states remain coherent

PEF Contribution

No belief timeline to “advance”

PEF maintains coexistence without decay

Legal Framing

- **PEF is not the invention itself**
- PEF is a **non-temporal substrate embodiment** that *supports* admissibility governance
- The spine explicitly states PEF is **optional and non-limiting**

✦ **This avoids unity problems** and prevents over-claiming consciousness or time.

2. Compositional Primitives Architecture Provisional

(Operator / mechanism invention)

What the Primitives Provisional Claims

- Explicit operators (WE, THEN, BECAUSE, WHILE, UNTIL, IF-THEN)
- Each operator has:
 - Preconditions
 - Activation rules
 - Update logic
 - Collapse conditions
- Collapse is governed, not heuristic
- Operators can fail or refuse to activate

Where It Maps into the Spine

Primary Spine Sections:

- §5.2 Admissibility Gate
- §5.3 Collapse Controller
- §7.1 Compositional Primitive Embodiment

How the Mapping Works

Spine Function	Primitive Contribution
Admissibility gate	Primitive preconditions
Constraint-forced collapse	Primitive collapse rules
Prohibition of heuristic resolution	Explicit operator activation
Deterministic governance	Operator semantics

Critical Point (this matters)

The primitives are **not the spine**.

They are **the enforcement machinery**.

The spine says *what must be governed*.

The primitives say *how governance is executed*.

✦ This is a textbook **means-plus-function relationship**, done safely.

3. Aurora Conceptual Blocks Provisional

(Structural / implementation invention)

What Aurora Blocks Claim

- Roles, Domains, Spans as structured state units
- Deterministic conceptual parsing
- Separation of structure from expression
- Auditable reasoning traces

Where It Maps into the Spine

Primary Spine Sections:

- §5.1 Epistemic State Representation
- §7.2 Structured State Embodiment
- §9 Advantages (auditability, determinism)

How the Mapping Works

Spine Requirement	Aurora Contribution
Explicit epistemic states	Roles/Domains/Spans

Spine Requirement Aurora Contribution

Parallel interpretations Multiple role bindings

Auditability Explicit state transitions

Compatibility with LLMs Expression layer separation

Legal Framing

- Aurora Blocks are a **reference implementation**
- They demonstrate enablement
- They are not required for scope

✦ This prevents “design-around by reimplementation” attacks.

4. Aurora + PEF Integration Provisional

(Continuity / anti-modularisation invention)

What the Integration Provisional Does

- Shows primitives operating within PEF
- Prevents separation of persistence from governance
- Establishes that coexistence is intentional, not accidental

Where It Maps into the Spine

Primary Spine Sections:

- §7 Embodiments (collectively)
- §10 Priority and Incorporation
- §3 Summary of the Invention (implicit support)

How the Mapping Works

This provisional:

- **Does not define a new invariant**
- **Does not expand scope**
- **Prevents hostile recombination**

